

National University of Computer and Emerging Sciences, Lahore Campus				
	Course:	Machine Learning for Robotics	Course Code:	CS 5096
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Date:	04-05-2026	Total Marks:	15
	Section:	A	Weight:	
	Exam:	Quiz 3	Roll No:	

Q1) Suppose we have 5 data points in a 2-dimensional space.

- a) Let the initial cluster centroids be $\mu_1=(2,2)$ and $\mu_2=(8,7)$. (5)

Calculate the Euclidean distance from each data point to both centroids and assign each data point to the nearest cluster.

x_i	$d(x, \mu_1)$	$d(x, \mu_2)$	Cluster (C_1/C_2)
(1, 3)			
(2, 5)			
(4, 4)			
(8, 2)			
(9, 4)			

- b) Compute the updated coordinates for the centroids. (2)

- c) A new data point $x_6=(3,4)$ is added to the dataset and assigned to Cluster C_1 . Calculate the coordinates of the updated centroid μ_1' including this new point. (2)

Q2) The discounted return at time t is given by

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \gamma^3 R_{t+4} + \dots$$

Suppose $\gamma = 0.5$ and the reward sequence is

$R_1 = 1, R_2 = 6, R_3 = 12, R_4 = 16$, then zero for R_5 and later.

a) Find G_3 and G_4 . (2)

b) A robot is navigating a 4x4 grid to find a goal while avoiding obstacles. The goal is a specific square in the grid where the agent receives a positive reward.

i) Define the state space. (2)

ii) Is the action space discrete or continuous? Justify your answer. (2)